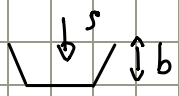
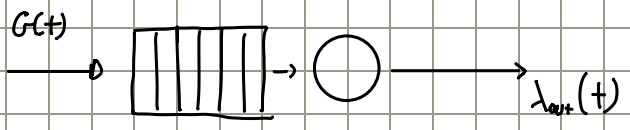
EXERCISES + NEW TOPIC

1) $G(t) = \dfrac{T/2 - \left| \text{rem}(t,T) - \frac{T}{2} \right|}{T/2} G_p$ (rem(t,T): reminder of division)

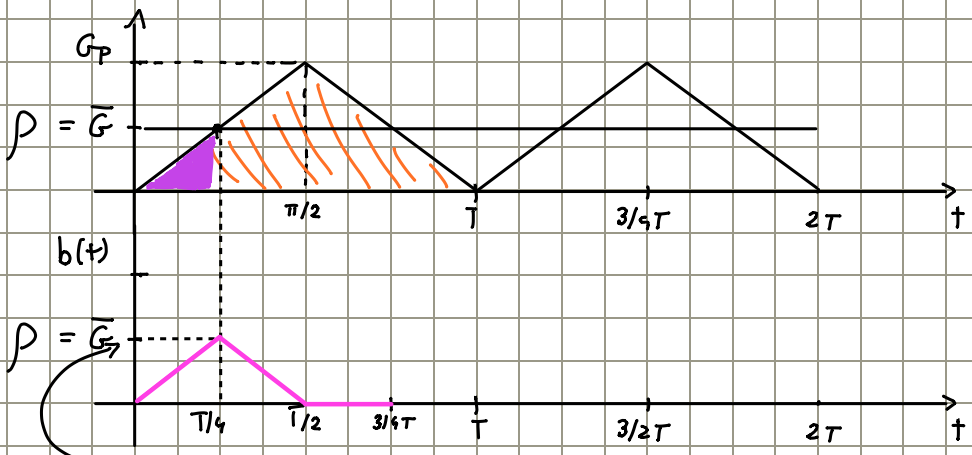




Find the $\bar{G} = \lim_{t\to\infty} \frac{1}{t} \int_0^t G(\sigma)\, d\sigma$

Set $\rho = \bar{G}$; $b(0) = 0$; $\lambda_{out}(t)$

$b = ?$



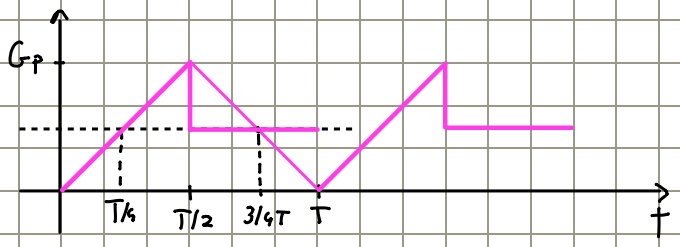
→ cosa è questo grafico?

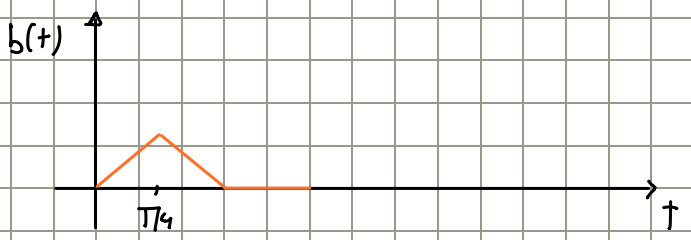
$$\bar{G} = \left(\frac{T \cdot G_p}{2} \right) \cdot \frac{1}{T} = \frac{G_p}{2}$$

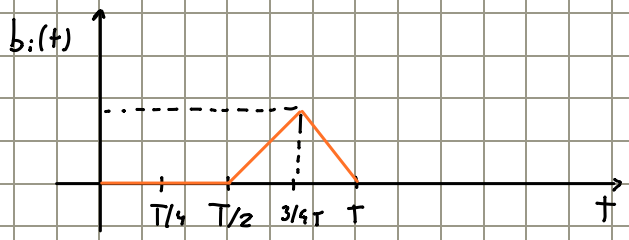
How do we expect λ_{out} to be like?

$$b(T/4) = \rho \cdot \frac{T}{4} - \frac{T}{4} \cdot \bar{G} \cdot \frac{1}{2} = \frac{\bar{G} \cdot T}{8} = \frac{G_p \cdot T}{16}$$

$$t \in (T/4, T/2] \rightarrow b(t) = \frac{G_p T}{8} - (G(t) - \rho)(t - T/4) \qquad t = T/2 \rightarrow b(T/2) = 0$$







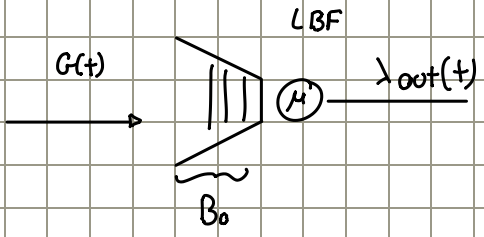
$\Rightarrow$ remains constant, why?

2) ON: CBR with $\lambda_p = 10$ Mb/s, $G(t) = \lambda_p$

$T_{ON} = 20$ ms

OFF: $G(t) = 0$

$T_{OFF} = 80$ ms

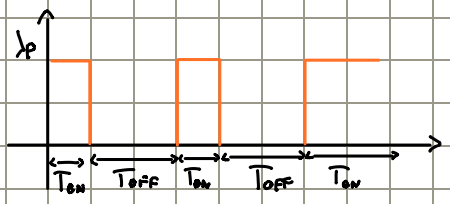


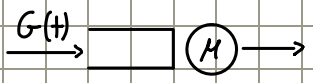
1. $\mu = ?$ so no traffic is discarded, for a sufficiently large B_0
2. $\mu = 4$ Mbps; $\sup_t b_0(t)$ = max buffer occupancy

   min B_0 to avoid losses
3. Max delay
4. $Pr[\lambda_{out}(t) = \mu]$

Answers:

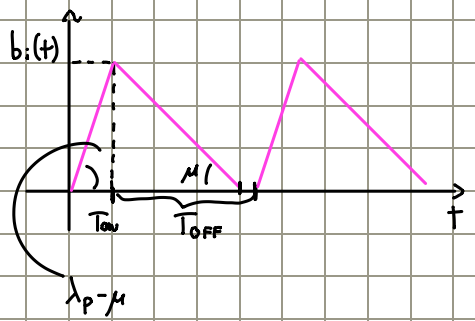
1.





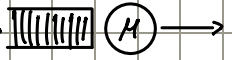
$$\mu \geq \overline{G} = \frac{G_{ON} \cdot T_{ON} + G_{OFF} T_{OFF}}{T_{ON} + T_{OFF}} = 2 \text{ Mbit/s}$$

2.



$$\sup_t b_0(t) = b_0(T_{ON}) = (\lambda_p - \mu) \cdot T_{ON} = 6 \cdot 10^6 \cdot 20 \cdot 10^{-3} = 120 \text{ Kbits}$$

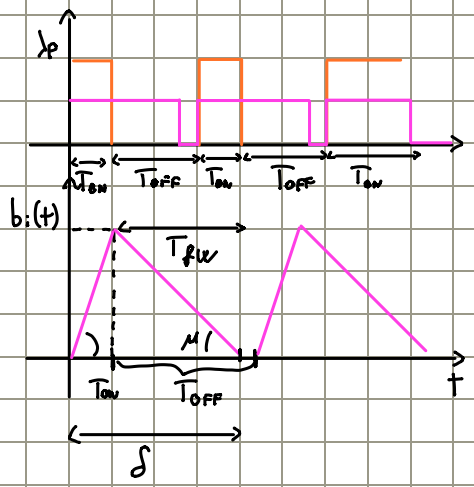
$B_0 \geq 120$ Kbits

3. Max delay, when buffer is full, $\xrightarrow{G(t)}$  $\xrightarrow{\mu}$

So max delay is time to get rid of data on the queue

$$d_{max} = \frac{b_i(T/2)}{\mu} \quad \text{(fluid model)}$$

4. $Pr[\lambda_{OUT}(t) = \mu]$



$b_i(t) > 0$ for $t \in (0, \beta)$ $\quad \delta = T_{ON} + T_{OFF}$

$$T_{fill} = \frac{b(T/2)}{\mu} = d_{max}$$

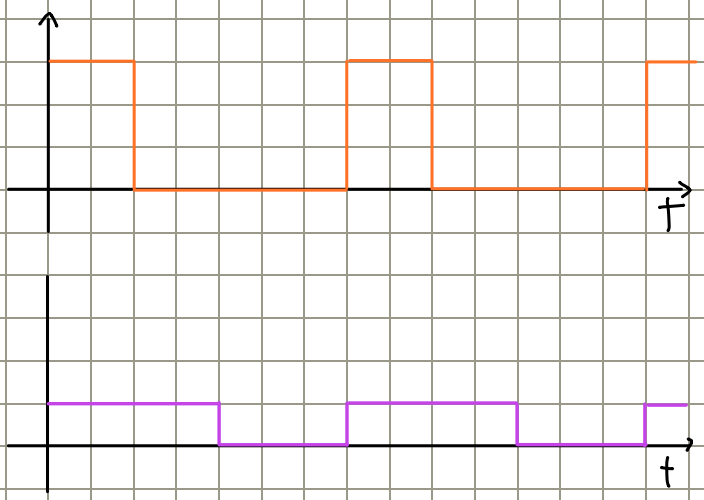
$$Pr[\lambda_{OUT}(t) = \mu] = \frac{\delta}{T_{ON} + T_{OFF}}$$

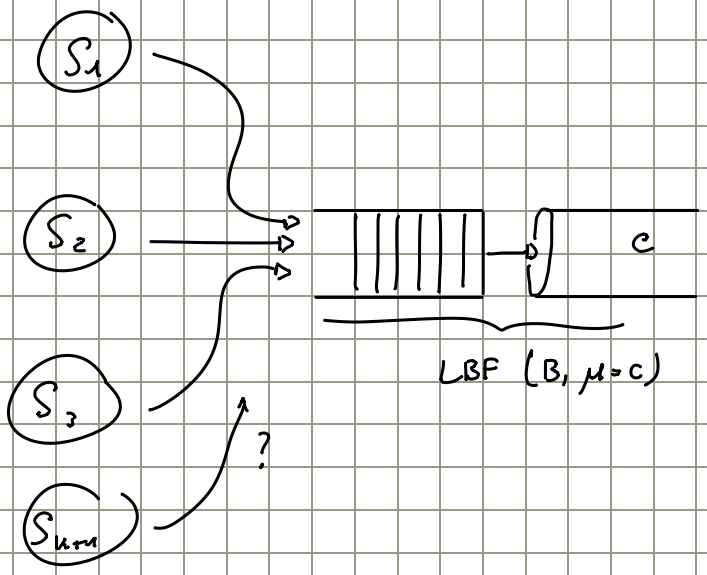
CALL - ADMISSION CONTROL (CAC)

Sources are ON-OFF processes

ON → CBR $G(t) = \lambda_p$, $T_{ON} =$

OFF → $G(t) = 0$ $T_{OFF} = const$





link with bitrate c

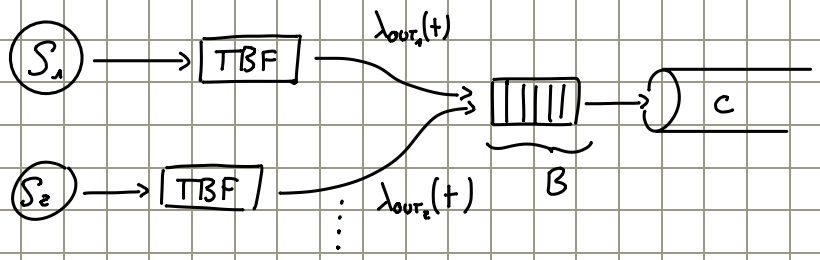
CAC
- Stochastic Bound → increase the efficiency, beter use of our resources
- Deterministic Bound

S_{n+1} is accepted iff $\sum_{j=1}^{n} \lambda_{p_j} \leq C$

Example $C = 10$ Mbit/s

$\lambda_{p_i} = 1$ Mbit/s $\Rightarrow \max \frac{C}{\lambda_p} = 10$ "calls"

$T_{ON} = 1$ ms ; $T_{OFF} = 1$ s



$G_{ON}(t) = \lambda_{p_i}$; T_{ON}

$G_{OFF}(t) = 0$; T_{OFF}

$\longrightarrow$ TBF$_j$

$$\rho_j = G_{ON} \cdot \frac{T_{ON}}{T_{ON} + T_{OFF}} = \overline{G}$$

$$b_j = (\lambda_{p_i} - \rho)\, T_{ON}$$

1. $\sum_{j=1}^{u} \rho_0 \leq C \rightarrow N_{rate} = \left\lfloor \frac{C}{\rho} \right\rfloor = 10000$
2. $\sum_{j=1}^{u} b_j \leq B \rightarrow N_{buffer} = \left\lfloor \frac{B}{b_i} \right\rfloor \cong 1000$

$B = 1$ Mbit

$$\rho = \overline{G} = 10^6 \frac{10^{-3}}{1 + 10^{-3}} \simeq 10^3 = 1 \text{ Kbps}$$

$$b = (\lambda_p - \rho)\, T_{ON} = (10^6 - 10^3) \cdot 20 \cdot 10^{-3} \cong 20 \text{ Kbits}$$